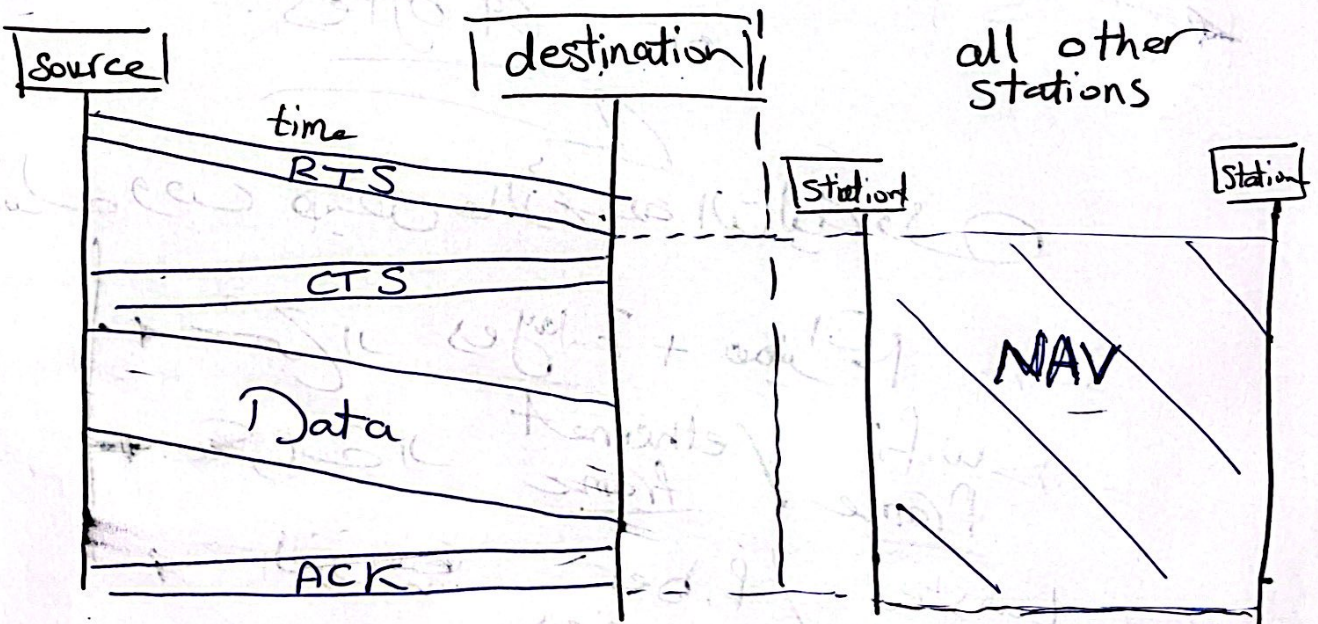
Revision

Midterm questions:-

1- a-wireless LANs cannot implement CSMA/CD state with the aid of the figures the relationship between CSMA/CA and NAV

→ Wireless LANs cannot implement CSMA/CD Because
→ 1- Collision May not be detected because of hidden station
2- for collision detection a station Must be able to sent data and Recieve collision signal at same time so this will have costly station and higher bandwidto

So Wireless LANs used CSMA/CA and NAV



1- the source sends RTS includes the duration time that need to occapy the channel

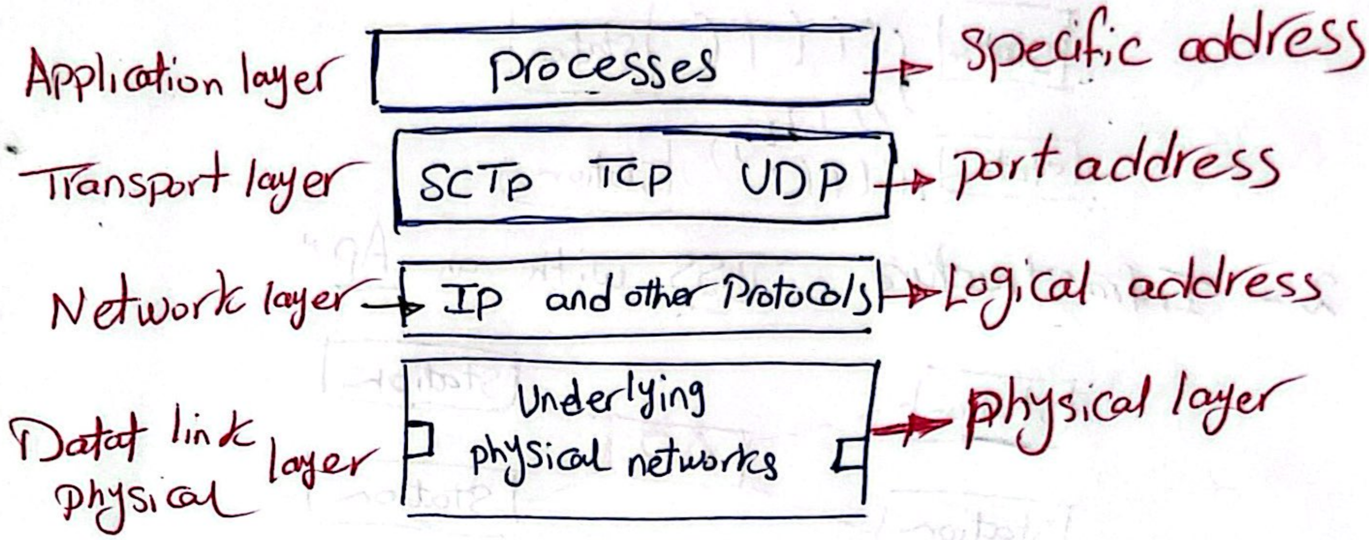
2- all other stations set to a timer called (NAV) at this time stations are not allowed to channel if empty)

3- each time station accesses the system and send RTS frame, other stations start (NAV)

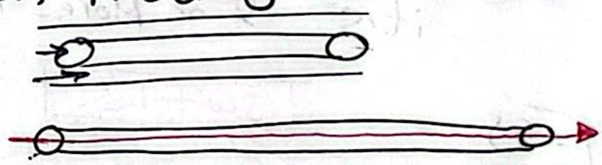
1

↳ Explain with aid of figures the Relationship of layers and addresses in TCP/IP --

TCP/IP has main 4 layes



↳ What is the Propagation modes at fibre optics ?



1- Single Mode :-

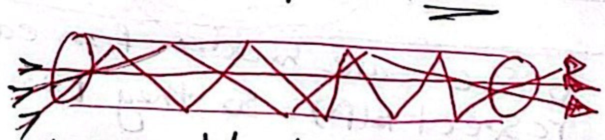
Highly focused source with thin core that send signal rapid by core have one signal at it in a time

2 - Multi Mode :-

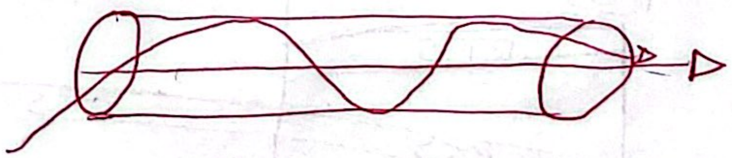
Multi beams from light source Move through the core in different paths --

and has 2 types

1- Multi Mode / step index

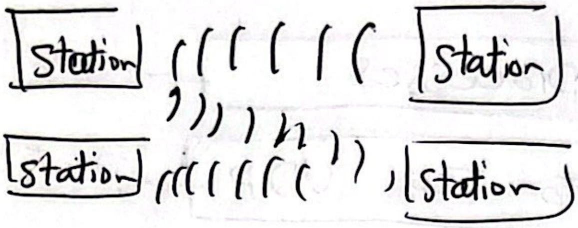


2- Multi-Mode / graded index

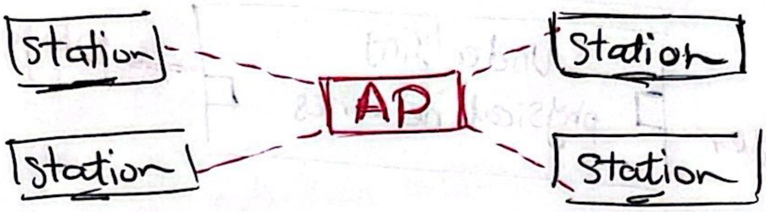


* What is the difference between " Ad-hoc - Infrastructure and ESS "

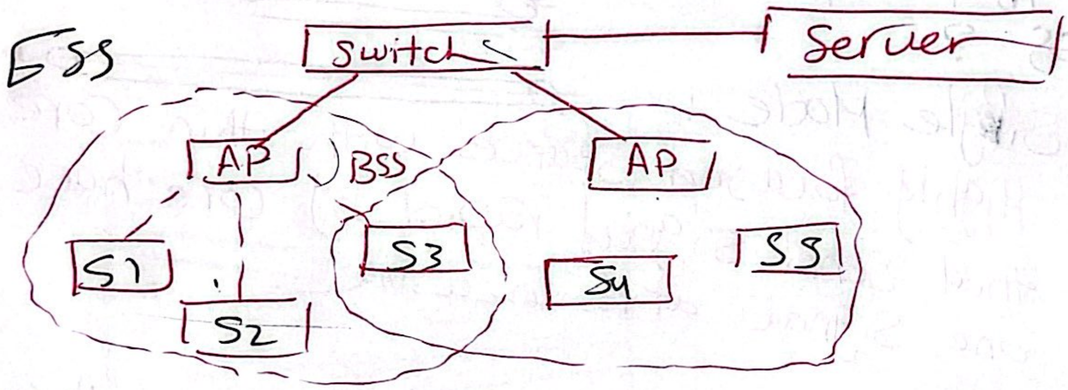
1- Ad-hoc Network :- it is "BSS without an AP"



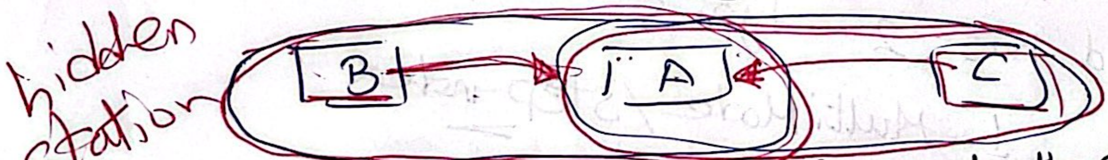
2 - Infrastructure :- "BSS with an Ap"



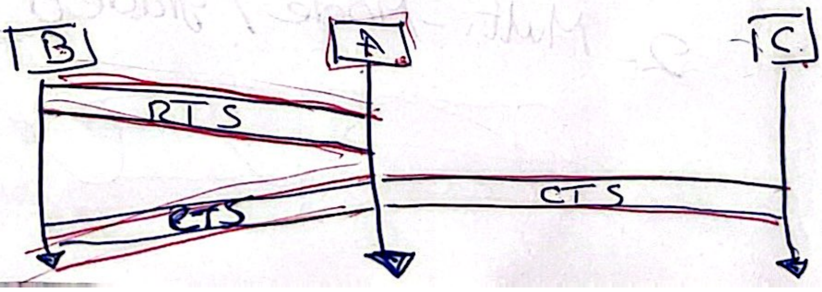
3- ESS (Extended service set"
it is 2 or More BSS with APS



➔ Show the problem of hidden station and how CSMA/CA avoid it by hand shaking :-



B & C are hidden from each other with respect to (A) so they two may send is same time



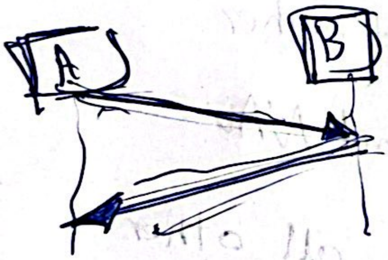
In the standard $\overset{10\,Mbps}{}$ Ethernet, if the Maximum Propagation time is $25{,}6\,\mu s$ what is the Minimum frame size?!

1- frame transmission time $= 2 * T_p = 51{,}2\,\mu s$ "worst case"

2- Minimum size of the frame is

$= $ Network speed $*$ frame transmission time

$= 10\,Mbps * 51{,}2\,\mu s$



$= 10 * 10^6 * 51{,}2 * 10^{-6}$

$= 512$ bits

or 64 bytes.

لوعندك وقت هيبص عالأسئلة التالية دي

شرح الـ 7-layers + مميزاتها

مكونات الـ wifi frame / ethernet frame

الفرق بين twisted pair / fiber optics

١- هيكون فيه سؤال سهل تحويله IP من Decimal لل Binary على الآلة الحاسبة أو العكس

٢- ذاكر كل class كويس جدًا واعرف ال Netid وال Host id

زي كدا $\boxed{}$. $\boxed{}$. $\boxed{}$. $\boxed{}$ class(A)

1byte ← Netid

Netid لازم كل binary يبدأ ب 0

عدد الاجهزة في كل شبكة = 2^{24} جهاز

عدد الشبكات = 2^7 = 128 شبكة

٣- الMaskid ← بتحل ال Netid كله بـ 1

Hostit كله بـ 0

يعني لو class(A) ← 255.0.0.0

٤- any IP address & Mask = Network id

any IP address || Mask complement = broadcast address

Example:- IP address 205.16.37.39/28

what is its first address, Last address, Number of addresses

Solution:-

لما يجلك /28 يبقى دا حجم ال Netid

لو عاوز ال first address حول لل Binary وخلي ال hostid بأصفار

Last address ب ...

first address

$\underbrace{11001101 \quad 00010000 \quad 00100101 \quad 0010}_{Netid}\ \underbrace{0111}_{host\ id}$

11001101 00010000 00100101 0010 0000

205 . 16 . 37 . 32

Scanned with CamScanner

Last address :-

11001101 00010000 00100101 0010 1111

205 . 16 . 37 . 47

Range = (Last address - first address) + 1

$$\begin{array}{r} 205 . 16 . 37 . 47 \\ - \quad 205 . 16 . 37 . 32 \\ \hline 0 . 0 . 0 . 15 \end{array}$$

addresses = 15 + 1 = 16 address

OR → No of address = $2^{\text{hostid bits}}$

$= 2^{32-28} = 2^4 = 16$

حل آخر ومهم :- نجيب ال Mask لل IP دا

و بما أن Netid = 28 و hostids 4

∴ Maskid = 11111111 11111111 11111111 11110000

first address = 11001101 00010000 00100101 00100111
And 11111111 11111111 11111111 11110000

11001101 00010000 00100101 0010 000

نفس الحل ← 205 . 16 . 37 . 32

Last address = 11001101 00010000 00100101 00100111
OR 0000 0000 0000 0000 00000000 00001111

11001101 00010000 00100101 0010 1111

205 . 16 . 37 . 47

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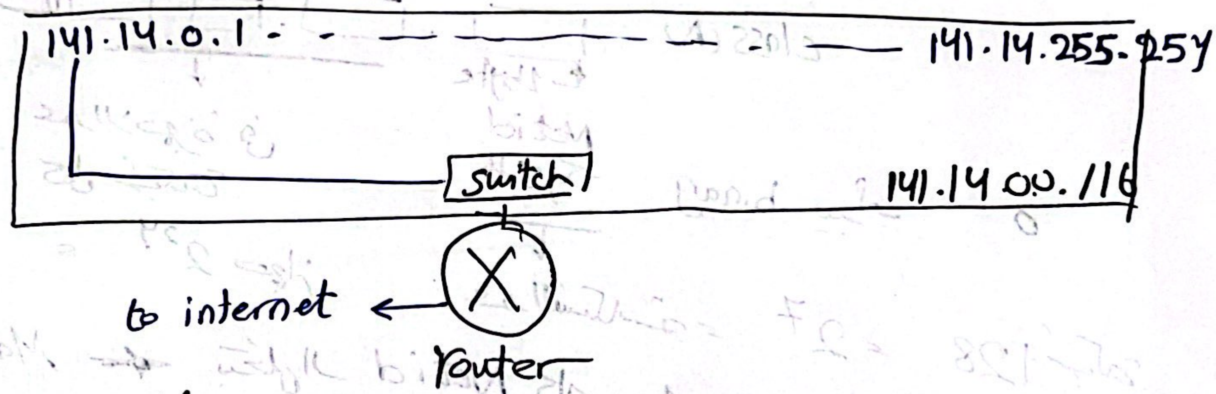
⊛ Network 141.14.0.0 /16 is required to subnetting for 4 equal subnets, show how and draw system before and after subnet --

1- **Before subnet** :-



⇒ all 2^{16} hosts are connected in same Network

⇐ عملية الـ Subnetting هجيب ٤ راوتر مع الراوتر الرئيسي بحيث الـ ٤ راوتر تقسم عليهم الـ 2^{16} عنوان المتاحين بحيث كل راوتر ياخذ 2^{14} عنوان يعني الربع

⇐ الفكرة أنك عشان تعمل كدا لازم عنوان لكل راوتر وعنوان بداية وعنوان نهاية

⇐ فهنستلف من Host id العنوان النت الفرعي الجديد وعشان محتاج ٤ عناوين هحتاج (2 bit) قبل ما كان الـ host id = 16 bit هاخذ اتنين ويبقى Net id = 18 bit و host id = 14 bit

⇐ أول شبكة نت (141 . 14 . 0 . 0 /18)

$$\underbrace{1000\ 1101\ .\ 0000\ 1110\ .\ 00}_{Net\ id}\underbrace{000000\ .\ 0000\ 0000}_{host\ id}$$

آخر عنوان فيها

1000 1101 0000 1110 0011 1111 1111 1111

141 . 14 . 63 . 255

141 . 14 . 64 . 0

الشبكة الثانية :-

أول عنوان 1000 1101 . 0000 1110 . 0100 0000 0000 0000

آخر عنوان 1000 1101 0000 1110 0111 1111 1111 1111

141 . 14 . 127 . 255

الشبكة الثالثة

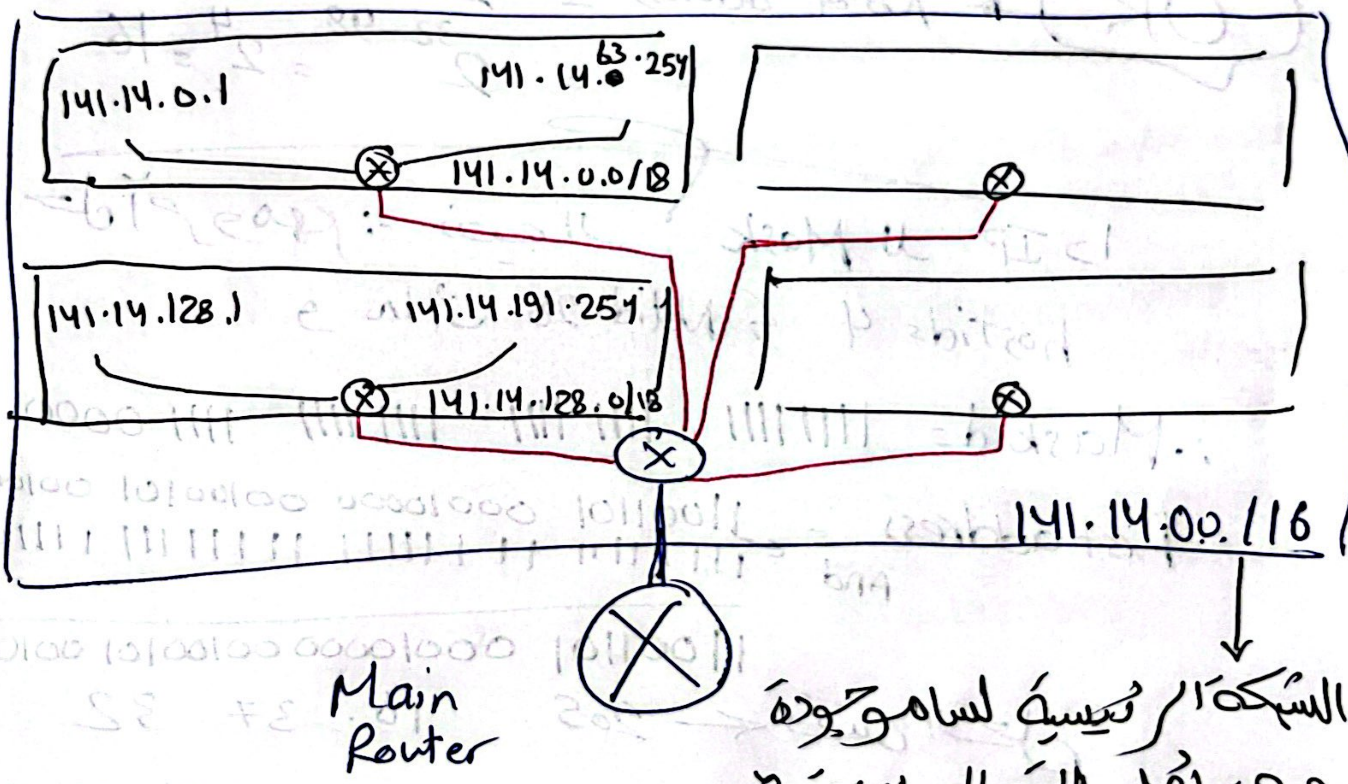
أول 141 . 14 . 128 . 0

آخر 141 . 14 . 191 . 255

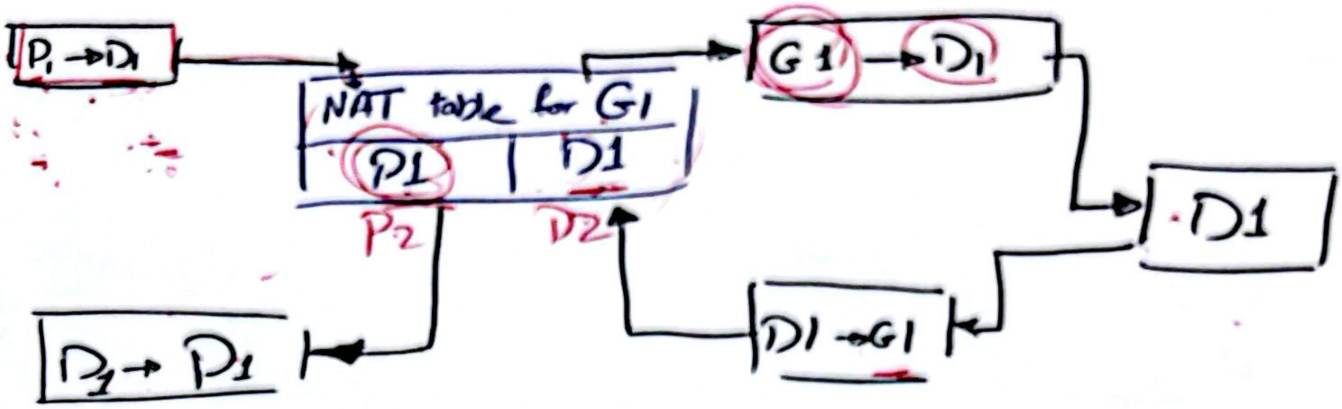
الشبكة الرابعة

أول 141 . 14 . 192 . 0

آخر 141 . 14 . 255 . 255



الشبكة الرئيسية لسا موجودة
وهي أصل الاتصال بالانترنت
ولكن أنت داخلياً قسمتها
لأربع شبكات



* شرح الصورة .

* Private Device (P1) inside a local network Sends a request to external destination D.1 Since Private address cannot be used directly the router performs NAT translation by mapping P1 to global Ip ⇒ G1

* the destination will replay to <u>G1</u> this will go to your router the router Perform NAT translation to map G1 into P1 and the replay will go to P1

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